01/12 Thermo syllabus

Monday, 12 January 2026 10:30 AM

Final project → Catalysis (Adsorption mechanism, kMC)

→ Capital → extensive } Convention.
letters (depend on the amount of substance) }

→ Postulate I (Duhem's) → n species; n+2 variables for bulk properties

→ Macro vs. Micro

↳ average behavior of length & time scales

Kinetic Theory of Gases

- Dilute (i.g. limit)
- non-interacting hard spheres
- Neutral atoms
- Polarizability of atom (induced dipole)

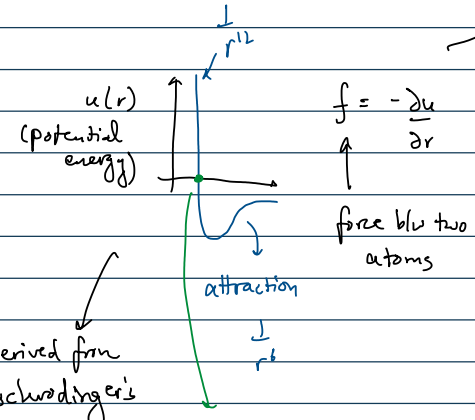
of molecules ↳ $N = nN_A$ (n: # of mols)

V → extensive volume

$\rho = \frac{N}{V}$ → no. density

$k = \frac{R}{N_A} = 1.38 e^{-23}$ J/K

$$\boxed{\rho = \frac{N}{V} = \frac{P}{kT}}$$



derived from schrodinger's eq

$f = -\frac{\partial u}{\partial r}$ ↑ force b/w two atoms

Coulomb's law still at play but these are averaged (so no two particles)

0 point ($u(r)=0$) gives $r=\sigma$ as effective size

6-12 Leonard Jones } very good model for noble gases (Xe, Ar, Kr)

gas → 10σ separation

liq → 1σ separation